

Theory Final—Sample Questions—Version of November 1, 2012

This is a collection of sample theory final questions. This is just to get some idea of the format of the final and the level of difficulty. These example questions do not guarantee anything about the topics of the questions on your final and they do not guarantee the exact level of difficulty of the questions on your final.

The actual final will consist of five or six questions of equal weight. Your lowest question score won't count.

Note: These questions were adapted from finals that used a different book. It has not been proofread as well as your final will be.

Procedures of the Theory Final

1. The only things that you are allowed to have out during the exam are your cheat sheet, pen(s), pencil(s), an eraser, a ruler, and a simple watch. Please note that you are not allowed to use cellphones, PDAs, etc. as a clock.
2. Please silence your cellphones etc.
3. Do not share your supplies with others and do not talk with others.
4. Use only scrap paper provided by me.
5. Please also hand in your scrap paper and cheat sheet when you are done.
6. Understanding the questions is part of the exam; you are not allowed to ask questions during the exam.
7. As usual, you only have to explain your answer if this is stated in the question.
8. If you use a theorem/construction from your book, please state what you are using where.
9. You may only leave the room after you finish your exam,
10. Your answers should be to the point.

1. Suppose that the alphabet is $\{a, b, c\}$.

For every class of languages in the table, you have to state whether the class is closed under complementation, concatenation, and intersection.

Write “yes” in a box if the class is closed under the operation. Write “no” if the class is not closed under the operation.

closed under	Complementation	Concatenation	Intersection
P			
The context-free languages			
The Turing-recognizable languages			
The decidable languages			
The regular languages			

- (a) For *one* of the boxes where you answered “no,” give an explicit example that shows why that particular class is not closed under that particular operation. (Clearly state which box you chose.)

- (b) For *one* of the boxes where you answered “yes,” informally argue why that particular class is closed under that particular operation. (Clearly state which box you chose.)

2. Consider the following grammar G :

$$S \rightarrow \epsilon \mid 01S \mid 10S \mid 0S1 \mid 1S0 \mid S01 \mid S10 \mid 1S \mid S1$$

(a) Show that this grammar is ambiguous.

At first glance, it may seem that the language generated by G is

$$L = \{x \in \{0, 1\}^* \mid n_0(x) \leq n_1(x)\}.$$

However, this is not true.

(b) For both of the following statements, circle “true” if the statement is true, and circle “false” if the statement is false.

i. $L \subseteq L(G)$ true / false

ii. $L(G) \subseteq L$ true / false

(c) Show that $L(G) \neq L$.

(d) Give a context-free grammar that generates L . (It suffices to give the rules.)

3. Let $L = \{a^i b^k c^i d^k \mid i, k \geq 0\}$.

(a) Explain informally why L doesn't seem to be a context-free language.

(b) Prove that L is not context-free.

4. Give the state diagram of a Turing machine that halts on all inputs and that accepts the language

$$\{x \in \{a, b\}^* \mid n_a(x) \neq n_b(x)\}$$

Give a short explanation to convince me that your Turing machine is correct.

5. A *2-PDA* is a like a PDA, except that it has two stacks.

(a) Give a language that can be accepted by a 2-PDA, but not by a PDA.

(b) Draw the state diagram of a 2-PDA that accepts your language from part (a). Explain the notation that you used in your state diagram.

(c) Give a *short, high-level* explanation to convince me that your 2-PDA is correct.

6. Consider the following language

$$E_{\text{TM}} = \{\langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset\}.$$

(a) Show that E_{TM} is undecidable.

(b) Is E_{TM} in RE (recursively enumerable), in coRE, in both, or in neither? Explain your answer.

(c) Show that the following language X is undecidable:

$$X = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are TM's and } L(M_1) \cap L(M_2) = \emptyset\}.$$

7. Consider the following decision problem.

$IS-OR-CLIQUE = \{ (G, k) \mid G \text{ is a graph, } k \text{ is an integer, and } G \text{ has a clique of size } k \text{ or an independent set of size } k \}.$

(a) Show that $IS-OR-CLIQUE$ is in NP.

(b) Give an explicit example that shows that the function f defined as $f(\langle G, k \rangle) = \langle G, k \rangle$ is *not* a polynomial-time many-one reduction from $INDEPENDENT-SET$ to $IS-OR-CLIQUE$.

(c) Show that $IS-OR-CLIQUE$ is NP-complete.